



HIGH YIELD WRAP-UP

And Q&A

SOME LOOSE ENDS

- **Peripheral Neuropathies**
- **Stroke**
- **Seizures and Epilepsy**

NEUROPATHY PATTERNS

Distal Symmetric Polyneuropathy – most common pattern, generalized peripheral nerve dysfunction affecting the longest nerves first. “Stocking/Glove distribution”

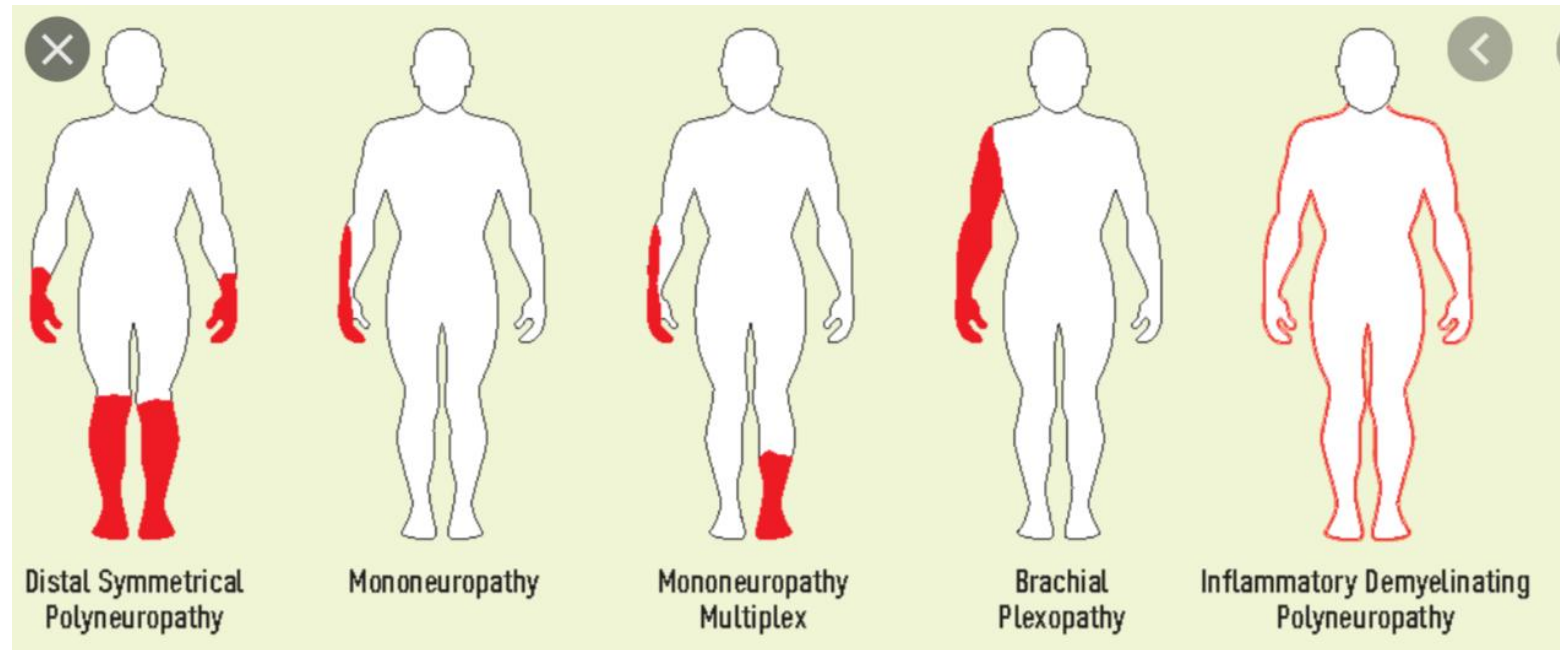
Radiculopathy – Symptoms confined to a single nerve root distribution. (Pinched nerve, slipped disc, etc)

Mononeuropathy – Symptoms confined to a single peripheral nerve. (Carpal Tunnel, Peroneal Neuropathy at fibular head, etc)

Mononeuritis Multiplex – multiple mononeuropathies.

NEUROPATHY PATHOLOGY

- Many different targets for injury in the peripheral nerve
- Most common is the myelin sheath
- The axons themselves can be directly injured
- The vascular supply can be disrupted (vasculitis)
- The cell body can be injured (neuronopathy)
- The nodes of Ranvier can be injured (nodopathy)



* Note that in some circumstances a radiculopathy can overlap closely with a mononeuropathy (like L5/peroneal)

NEUROPATHY DISTRIBUTIONS

GUILLAIN-BARRE SYNDROME

- Also called acute inflammatory demyelinating polyneuropathy
- Presents as acute-subacute onset of *ascending* paralysis and sensory disturbance
- Associated with recent illness, the most common is campylobacter
- Treat with IVIg or Plasma exchange

BELL'S PALSY

- Idiopathic facial nerve palsy
- Associated with hyperacusis and taste sensation loss (from facial nerve branch supplying the stapedius and the chorda tympani)
- DOES NOT SPARE THE FOREHEAD, causes eye closure weakness
- Treat with course of corticosteroids, only add antiviral like acyclovir if patient has fevers or rash
- Frequently improves nearly back to baseline, though in patient's that do not improve over several month, imaging should be obtained to rule out space occupying lesion.
- Exam and history should focus on finding any CNS findings that might sway you towards a more central localization.
- Parotid tumors can sometimes masquerade as a Bell's palsy so palpate the parotid for masses.

STROKE SYNDROMES

A 48 year old man with a history of atrial fibrillation presents with sudden onset of right facial droop, inability to speak, right arm and leg weakness, and leftward eye deviation.

1. What part of the brain would you localize these symptoms to?
2. Which blood vessels contribute to this area?
3. Does his medical history suggest a particular etiology?

STROKE SYNDROMES

A 92 year old man with a history of hypertension, diabetes, obstructive sleep apnea presents with complaints of left sided weakness after waking up from a nap. On exam you find that he is unable to lift the left leg antigravity, but retains antigravity strength in the left upper extremity. Sensation is similarly diminished in the lower extremity but mostly intact in the upper.

1. What part of the brain would you localize these symptoms to?
2. Which blood vessels contribute to this area?

STROKES STROKES STROKES

A 62 year old woman presents after she had a small car accident, hitting several parked cars outside of the mall. On examination she has a right homonymous hemianopia but no other significant neurologic deficits. If this is a cortical stroke...

1. What part of the brain would you localize these symptoms to?
2. Which blood vessels contribute to this area?
3. Any other localizations for this isolated deficit (if non cortical?)

STROKE? YES, MORE STROKE

45 year old man with hx of hypertension, diabetes, osa, 20 pack year smoking history presents for evaluation of sudden onset left face, arm and leg weakness, as well as double vision (right eye down and out) and fixed, dilated right pupil.

1. What part of the brain would you localize these symptoms to?
2. Which blood vessels contribute to this area?
3. Does his medical history suggest a particular etiology?

STROKE? MAYBE. PROBABLY.

You are seeing a 50 year old man in the ER who has complaints of extreme pain throughout his entire right body and face. This began over the past week. About 2 weeks ago he was diagnosed with CVA and treated with tpa, which was only partially successful. His presenting symptom at that time was right sided sensory loss.

1. What part of the brain would you localize these symptoms to?
2. Which blood vessels contribute to this area?

CAN'T BE MORE STROKES

It's a
stroke
though

65 yo m w all the vascular risk factors presents with sudden onset vertigo, difficulty walking, dysarthria, leftward tongue deviation, right sided pain and temperature sensation loss of the body, left sided pain and temperature loss on the face, and left sided ptosis and constricted pupil.

1. What part of the brain would you localize these symptoms to?
2. Which blood vessels contribute to this area?
3. Does his medical history suggest a particular etiology?

EPILEPSY HIGH YIELD

Seizure: uncontrolled, synchronized neuronal firing

Epilepsy: a disease of recurrent seizures

Seizures can be:

- * Focal**
- * Awareness impaired or Awareness retained**
- * Generalized**

GENERALIZED SEIZURES

Absence seizures – impaired consciousness without convulsions. 3hz spike-wave on EEG.

Tonic-Clonic seizure – the full 'grand-mal' seizure with initial stiffening followed by generalized, synchronous convulsions. Post-ictal period present.

Status Epilepticus – continuous or recurrent seizures. Any seizure lasting longer than 5 minutes or recurrent seizures without inter-ictal return to baseline.

COMMON CAUSES BY AGE

Adolescents (<18 years old): Trauma, genetic epilepsies, substance abuse, tumors.

Young Adults (18-35): Trauma, Alcohol withdrawal, substance abuse, tumors, autoimmune encephalitis.

Adults and Elderly (>35 years): **Strokes**, tumors, alcohol withdrawal, neurodegenerative disease, autoimmune encephalitis.

SIMPLE FEBRILE SEIZURES

- 6 months to 5 years age
- Single generalized seizure with fever
- Brief 2-3 minutes
- NONFOCAL
- May carry slight increase risk for epilepsy but not high

COMPLEX FEBRILE SEIZURES

- More than one event in 24 hours
- Longer than 15 minutes
- Focal symptoms
- Associated with greater risk for epilepsy
- More likely to have developmental delay

INFANTILE SPASMS

- Generalized epilepsy with onset in infancy
- Brief flexion or extension movements
- EEG pattern is called Hypsarrhythmia – a chaotic, giant slow waves, multifocal spikes and sharp waves.
- West syndrome refers to the epilepsy syndrome of infantile spasms
- Global developmental delay
- Treat with ACTH or corticosteroids

BENIGN EPILEPSY WITH CENTROTEMPORAL SPIKES

- Most common childhood epilepsy syndrome
- Peak around 7-8 years old
- Focal seizures, hemiconvulsions, dysarthria, paresthesia of tongue, lips, gum and cheek.
- Usually occur just before sleep or after awakening
- Developmentally normal
- EEG shows classic pattern of centrotemporal spikes
- Great prognosis, AEDs infrequently needed, 99.8% remission by age 18

JUVENILE MYOCLONIC EPILEPSY (JME)

- Begins in adolescence, normal birth and development
- Normal exam, normal intelligence
- GTC seizure often occur upon awakening
- Occasional myoclonic jerks (quick involuntary limb jerks) w fatigue
- Does not impair intelligence
- Treat with Valproic Acid
- EEG shows 4-6 Hz Spike-and-wave discharges

STUDY TIPS

- This is not an exhaustive guide, please continue to review your study material
- These high yield topics are frequently tested on the boards.
- Making some of these connections (4-6 hz spike+wave = JME in the right setting) can be valuable for boards strategies but not for clinical life as much.
- Good luck!